CASH TO COMPANY

# Item Bank

*"Entrepreneurship is organized problem-solving"*

Per-unit quiz / test items · 4 units · ETS-quality alignment to Jump$tart, CEE, CCSS, C3, CASEL, NGSS

# Grade 9 — Cash to Company Item Bank

**Year theme:** *Entrepreneurship is organized problem-solving.* This bank covers all four Grade 9 units (Map the model · Discover the customer · Model the money · Test the bet). Items use BMC, customer-discovery, P&L, and MVP-test scenario style. Distractors target the Grade 9 Common Misconceptions table. No family-money probes; "for a hypothetical" framing throughout. Cash's Grade 9 voice: "*Being wrong on Day 1 is the deal.*"

## Unit 1: Map the model

### MC1. "On a Business Model Canvas, every box should start as..."

A. Confirmed truth · B. An assumption flagged confirmed / hypothesis / imagined · C. A bullet list of features · D. A question

**Answer:** B · **Standard:** *NGSS-ETS HS-ETS1-2 — Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.* · **Strand outcome (BAVC, 9):** *I can map a Business Model Canvas for a real venture (9 boxes).* · **Misconception targeted:** "*Every BMC box needs to be perfect on Day 1*" — corrects.

### MC2. "The 9 boxes of a BMC include all EXCEPT..."

A. Customer Segments · B. Value Proposition · C. Marketing Calendar · D. Cost Structure

**Answer:** C · **Standard:** *Jump$tart Standard 1 (Spending and Saving), Benchmark 12-1 — Develop a plan for spending and saving.* · **Strand outcome (BAVC, 9):** *I can map a Business Model Canvas for a real venture.* · **Misconception targeted:** "*BMC = marketing plan*" — corrects.

### MC3. "Marginal-benefit / marginal-cost reasoning is most useful in BMC for..."

A. Designing the logo · B. Deciding 'what would have to be true' for an MVP go/no-go · C. Hiring · D. Naming the company

**Answer:** B · **Standard:** *C3 D2.Eco.2.9-12 — Use marginal benefits and marginal costs to construct an argument.* · **Strand outcome (LRCI, 9):** *I can run a 2-week MVP test with a defined "what would have to be true" success metric, and write a kill/iterate/scale recommendation.* · **Misconception targeted:** "*Decisions are about feeling*" — corrects.

### SA1. "Build a v1 BMC for a hypothetical 'protein bar for soccer players' venture. Fill all 9 boxes; flag each as confirmed / hypothesis / imagined."

**Model answer:** "*1. Customer Segments: 7th-9th-grade soccer players (HYPOTHESIS — 5 interviews suggest, not confirmed at scale). 2. Value Prop: portable savory snack on game days (HYPOTHESIS). 3. Channels: team group chats + locker hallway (IMAGINED). 4. Customer Relationships: pre-order Friday (IMAGINED). 5. Revenue Streams: $1.75/bar (HYPOTHESIS — 3 of 5 said yes). 6. Key Activities: source ingredients, assemble, deliver (CONFIRMED — small batch tested). 7. Key Resources: kitchen, ingredients, labels (CONFIRMED). 8. Key Partners: local nut wholesaler (HYPOTHESIS). 9. Cost Structure: $0.85/bar materials + packaging (CONFIRMED via prototype).*"

**Scoring guide (3-2-1):** - **3:** All 9 boxes filled, all flagged with confidence label, at least 3 marked HYPOTHESIS or IMAGINED. - **2:** 8 of 9 boxes, all flagged. - **1:** Boxes filled but flags missing.

### SA2. "Of your 9 BMC boxes, which 2 are most likely to be wrong, and why? What evidence would change your mind?"

**Model answer:** "*1. Channel ('locker hallway') — IMAGINED, no real test; if hallway-stop rate is below 10%, channel needs to be replaced with team group-chat pre-order. Evidence to change my mind: 5-day stop count below threshold. 2. Price ($1.75) — only 3 of 5 said yes; if conversion drops below 50% in real money-down test, drop to $1.50. Evidence: actual pre-order data, not stated preference.*"

**Scoring guide (3-2-1):** - **3:** Names two specific boxes AND the evidence threshold for each. - **2:** Names two boxes, evidence vague. - **1:** Names one or no thresholds.

### Performance Task: BMC v1 with Confidence Flags (90 min).

Child completes all 9 BMC boxes for their Capstone venture, flags each as confirmed / hypothesis / imagined, writes a 1-paragraph "the 2 boxes I'm least sure about" memo with falsifiable evidence thresholds. **Rubric:** Business Plan Rubric (Grade 9): (1) all 9 boxes filled, (2) flags applied correctly, (3) least-sure memo names threshold evidence. **Strand outcome:** BAVC.

### Extension (advanced learner):

"*Your BMC has 6 'imagined' boxes. Design a 2-week sequence of cheap tests, one per box, that would convert each from imagined to hypothesis or confirmed.*" Anchors test-design discipline.

# Unit 2: Discover the customer

### MC1. "The 'Mom Test' rule for customer interviews is..."

A. Bring your mom · B. Ask about THEIR life, not your idea · C. Use a 5-point scale · D. Record audio always

**Answer:** B · **Standard:** *CCSS-ELA W.9-10.7 — Conduct research and integrate technical information.* · **Strand outcome (PSIG, 9):** *I can run a customer-discovery cycle with 5 real interviews (or recorded surrogates), code the patterns, and write what changed in my hypothesis.* · **Misconception targeted:** "*Customer interviews are sales pitches*" — corrects.

### MC2. "After 5 customer interviews, your hypothesis hasn't changed. The most likely reason is..."

A. Your hypothesis was correct · B. Your interviews were leading or confirmation-biased · C. The customers were wrong · D. 5 interviews is enough

**Answer:** B · **Standard:** *CCSS-Math S-IC.B.4 — Estimate from a sample.* · **Strand outcome (PSIG, 9):** *I can run a customer-discovery cycle with 5 real interviews and write what changed in my hypothesis.* · **Misconception targeted:** "*If I do customer discovery, I'll find the truth*" — corrects.

### MC3. "Customer discovery surfaces..."

A. Absolute truths · B. Patterns and disconfirmations · C. Final pricing · D. Product specs

**Answer:** B · **Standard:** *CCSS-ELA SL.9-10.4 — Present findings clearly and concisely.* · **Strand outcome (PSIG, 9):** *I can run a customer-discovery cycle with 5 real interviews.* · **Misconception targeted:** "*Discovery = certainty*" — corrects.

### SA1. "You ran 5 interviews. Code the pattern: name the top 2 customer pains AND quote one customer per pain in their words."

**Model answer:** "*Pain 1: bar packaging breaks in soccer bag. 'I had peanut butter all over my cleats — never again.' (Sam, 8th-grader.) Pain 2: 'sweet bars feel weird before a game.' 'I want salty when I'm sweating, not chocolate.' (Rosa, 9th-grader.)*"

**Scoring guide (3-2-1):** - **3:** Two pains AND a real-sounding customer quote per pain. - **2:** Two pains, one quote. - **1:** Pains listed without quotes.

### SA2. "Write what changed in your hypothesis between BMC v1 and v2. Two changes minimum, each with the interview evidence that drove it."

**Model answer:** "*v1 hypothesis: 'sweet bar at $3.' v2: 'savory bar at $1.75.' Change 1: drove savory because 4 of 5 interviews described 'sweet feels weird before a game' (Rosa, Sam, Aiden, Ben). Change 2: drove $1.75 because price-test interview*

*('$3 is too much; $1.50-$2 feels right') from 3 of 5 set a clear willingness-to-pay ceiling.*"

**Scoring guide (3-2-1):** - **3:** Two changes with interview-evidence per change. - **2:** Two changes, one with weak evidence. - **1:** One change.

### Performance Task: Customer Discovery Pack (5 interviews + memo + Canvas v2; 1 week deployment).

Child conducts 5 real interviews, codes patterns, writes a memo (top 2 pains, top 1 surprise), and produces BMC v2 with at least 2 specific revisions. **Rubric:** Customer Research Rubric (Grade 9): (1) 5 interviews documented with quotes, (2) pattern memo identifies pains in customer language, (3) BMC v2 has ≥2 evidence-driven revisions. **Strand outcome:** PSIG + CBM.

### Extension (advanced learner):

"*One of your 5 interviews directly contradicts the other 4. Walk through how you decide whether the outlier signal matters. (Hint: is the outlier a real customer with money, or a polite respondent?)*" Targets evidence-weighting.

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## Unit 3: Model the money

### MC1. "In a 1-year P&L, gross margin =..."

A. Revenue - Operating Expenses · B. Revenue - COGS · C. Revenue - Net Income · D. Revenue × Margin %

**Answer:** B · **Standard:** *CCSS-Math A-CED.A.2 — Create equations in two or more variables.* · **Strand outcome (MFL, 9):** *I can build a simple one-year P&L (income statement) for a venture with revenue, COGS, gross margin, operating expenses, and net income.* · **Misconception targeted:** "*Margin = profit*" — corrects.

### MC2. "Net Income on a P&L = Revenue – COGS – Operating Expenses. If revenue is $20,000, COGS $8,000, operating expenses $7,000, what is net income?"

A. $20,000 · B. $5,000 · C. $12,000 · D. $13,000

**Answer:** B · **Standard:** *CCSS-Math A-CED.A.2.* · **Strand outcome (MFL, 9):** *I can build a simple one-year P&L for a venture.* · **Misconception targeted:** "*Net = revenue minus one cost*" — C and D bait this.

### MC3. "A P&L tells you..."

A. Whether the math works · B. Whether you'll stay alive cash-wise · C. Both equally · D. Your tax bill

**Answer:** A · **Standard:** *CEE Standard 1 (Earning Income), Grade 12 — Entrepreneurs accept the risks in organizing resources to produce goods and services and hope to*

*earn profits.* · **Strand outcome (MFL, 9):** *I can build a simple one-year P&L.* · **Misconception targeted:** "*P&L tells me if my business is healthy*" — corrects (cash flow is Grade 11).

### SA1. "Build a 1-year P&L for: $25,000 revenue, $9,000 COGS, $5,500 operating expenses (rent, marketing, packaging). Show gross margin, gross margin %, and net income."

**Model answer:** "*Revenue: $25,000. COGS: $9,000. Gross Margin: $16,000. Gross Margin %: 64%. Operating Expenses: $5,500. Net Income: $10,500.*"

**Scoring guide (3-2-1):** - **3:** All five line items correct. - **2:** GM right, % rounded slightly off. - **1:** Major errors.

### SA2. "Run a 2-scenario sensitivity on the P&L above: (a) revenue drops 20%; (b) COGS rises 15%. What's net income in each? Which is the bigger threat?"

**Model answer:** "*(a) Revenue $20,000, COGS $9,000, GM $11,000, OpEx $5,500, NI = $5,500. (b) Revenue $25,000, COGS $10,350, GM $14,650, OpEx $5,500, NI = $9,150. The 20% revenue drop is the bigger threat (NI falls 48% vs. 13%); the venture is more revenue-sensitive than COGS-sensitive at these levels.*"

**Scoring guide (3-2-1):** - **3:** Both scenarios computed correctly AND the comparative interpretation named. - **2:** Both computed, interpretation vague. - **1:** One scenario right.

### Performance Task: 1-Year P&L Pack (90 min over 2 sessions).

Child builds a complete 1-year P&L (revenue model, COGS, gross margin, operating expenses, net income) AND a 2-scenario sensitivity table for their Capstone venture. **Rubric:** Budget Project Rubric (Grade 9): (1) all P&L lines present, (2) math correct, (3) sensitivity 2-scenario, (4) interpretation names sensitivity. **Strand outcome:** MFL.

### Extension (advanced learner):

"*Your P&L shows $10,500 net income but you're afraid of running out of cash by month 4. Why might this happen?*" Names the profit-vs-cash gap; previews Grade 11 cash flow.

## Unit 4: Test the bet

### MC1. "An MVP is..."

A. The launch version · B. The smallest version that could change your mind · C. A prototype with all features · D. A product spec

**Answer:** B · **Standard:** *NGSS-ETS HS-ETS1-2.* · **Strand outcome (LRCI, 9):** *I can run a 2-week MVP test with a defined "what would have to be true" success metric,*

*and write a kill/iterate/scale recommendation.* · **Misconception targeted:** "*MVP means launch-ready*" — corrects.

### MC2. "A 'what would have to be true' success metric is..."

A. A wish · B. A pre-committed threshold (e.g., '15 paying customers in 14 days') that distinguishes kill/iterate/scale · C. A revenue target · D. A vibe check

**Answer:** B · **Standard:** *C3 D2.Eco.2.9-12 — Use marginal benefits and marginal costs to construct an argument.* · **Strand outcome (LRCI, 9):** *I can run a 2-week MVP test with a defined "what would have to be true" success metric.* · **Misconception targeted:** "*MVP success = vibes*" — corrects.

### MC3. "You ran a 2-week MVP test. Result: 8 paying customers (threshold was 15). Recommendation?"

A. Scale aggressively · B. Iterate or kill — depending on what you learned about WHY the threshold missed · C. Scale conservatively · D. Run a 3rd week

**Answer:** B · **Standard:** *CASEL Responsible Decision-Making.* · **Strand outcome (LRCI, 9):** *I can run a 2-week MVP test.* · **Misconception targeted:** "*Failed MVP = quit*" — corrects.

### SA1. "Define a 'what would have to be true' success metric for a hypothetical 2-week MVP of a $1.75 protein-bar venture. Pre-commit a kill/iterate/scale rule."

**Model answer:** "*WWHTBT: 20 paying customers AND 30%+ repeat-purchase rate within 14 days. Kill: <10 paying customers OR <10% repeat. Iterate: 10-19 paying customers OR 10-29% repeat — change one variable (price, channel, or product spec) and re-run. Scale: ≥20 paying customers AND ≥30% repeat — increase batch size 3x for next 2 weeks.*"

**Scoring guide (3-2-1):** - **3:** Specific metric AND specific kill/iterate/scale thresholds. - **2:** Metric specific, thresholds vague. - **1:** Metric vague.

### SA2. "Your MVP test ended at 8 paying customers (threshold 20). Write a kill / iterate / scale recommendation in 1 paragraph with reasoning."

**Model answer:** "*Recommend ITERATE. Below kill threshold but interview data suggests the failure was channel (only 25% of soccer-team group chat saw the post), not product. Iteration plan: replace 'team chat post' with in-person Friday locker drop for 2 weeks, hold price/product constant. Re-test the same metric. If next 2 weeks miss 15+ paying, kill — that signals product-market fit, not channel, is the issue.*"

**Scoring guide (3-2-1):** - **3:** Recommendation + reason + specific iteration plan + re-test threshold. - **2:** Recommendation + reason, no specific iteration. - **1:** "Try again."

### Performance Task: 2-Week MVP Test (2 weeks deploy + 1-page memo).

Child runs an actual 2-week MVP test with a pre-committed WWHTBT metric, tracks results daily, and writes a kill/iterate/scale recommendation grounded in evidence.

**Rubric:** Reflection Writing Rubric (Grade 9): (1) WWHTBT metric pre-committed BEFORE test, (2) data tracked daily, (3) recommendation matches data + names iteration variable. **Strand outcome:** LRCI + PSIG.

### Extension (advanced learner):

"*Your MVP succeeded at the threshold — 22 paying customers in 14 days. Before scaling, what 2 second-order risks should you stress-test?*" Targets pre-scale checks; previews Grade 10 premortem.

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**Total this grade:** 12 MC + 8 SA + 4 Performance Tasks + 4 Extensions = 28 items.